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A user experience evaluation of a mobile application for disseminating site-specific impact-based flood warnings: The A4alerts app

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Abstract

People-centred early warning systems must ensure that their warnings are timely, actionable and get to the largest number of exposed individuals in the shortest time possible. Due to the increased access to high-bandwidth communication, mobile applications have become a valuable complement to traditional sources of emergency information. However, there is limited research on how useful end-users perceive these apps and their functionalities in the context of flood emergencies. This research presents the first online user experience evaluation of the A4alerts, a crisis app developed to disseminate the flood warnings triggered by the operational impact-based early warning system in Catalonia, Spain. Following an interactive experiment using the A4alerts app to access emergency information within real-based flood scenarios, participants were prompted to measure their experience using statements linked to criteria on usefulness, functional appropriateness, and adoption. The evaluation results suggest that sampled participants considered the A4alerts a useful communication tool to support end-users and their emergency actions. In particular, the local impact-based flood warnings, regional official warnings and self-protection action checklist were ranked as the most useful functionalities implemented within the app. These findings justify future iterations of the A4alerts for supporting the rapid dissemination of emergency information and helping communities reduce the impacts of flood events.

KEYWORDS

emergency management, flash flood, forecasting and warning, risk communication

1 | INTRODUCTION

Flash floods result from extreme precipitation in an area exceeding the natural or drainage system capacities (Seneviratne et al., 2021). Typically, these affect small to medium size catchments characterised by steep topography, thus, prone to convective storms (Corral et al., 2019; Ritter et al., 2020). As seen during the 2021 floods in Belgium, Germany, and The Netherlands, flash floods can have devastating negative socio-economic impacts in communities, including grave danger to human lives and economic damages in the range of

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billion euros for a single event (Netzel et al., 2021). Regardless of warnings issued in advance by the national meteorological and hydrological services for the 2021 floods, the message on the potential impacts and mitigation actions failed to reach some of the most vulnerable communities (Fekete & Sandholz, 2021). These previous events have exposed the significant gap between the impact forecasting tools available and the timely warnings communities need to effectively mitigate the negative consequences of floods. Alarmingly, these types of extreme floods are only becoming more frequent, with most of the recorded events from 1860 and 2016 in Europe categorised as flash floods (Bednar-Friedl et al., 2022). Hence, due to their explosive and rapidly evolving nature (usually not more than a few hours), integrating response and effective risk communication mechanisms to support communities during emergencies is essential and a major challenge Aligned with target G of the Sendai Framework for Disaster Risk Reduction, end-to-end early warning systems (EWS) must incorporate the following four interrelated pillars: (i) disaster risk knowledge; (ii) detection, monitoring, analysis, and forecasting; (iii) dissemination and communication; and (iv) preparedness at all levels. By integrating the above key elements, EWS can increase the likelihood of issuing warnings that are understandable, actionable and get to the largest number of exposed individuals in the shortest time possible (UNISDR, 2015). Communities expect hazard and local impact information to be readily available and accessible, especially when the time window for action during flash flood emergencies is limited (Dallo & Marti, 2021).

Information and communication technologies (ICT) have become promising tools to support the rapid dissemination of flood warnings; process and share real-time emergency information; and establish communication channels between authorities, responders and citizens during evolving flood emergencies (Grinko et al., 2019). While the traditional methods of dissemination (e.g., TV news and radio) remain essential sources of information, the increasing number of people using wireless technologies with accessible internet connection allows the exploration of more flexible and innovative methods for flood risk communication (Stanciugelu et al., 2017). Within crisis-related ICT, mobile hazard crisis apps started to be officially implemented in parallel to help facilitate the swift distribution of context-specific messages (Reuter, Kaufhold, & Knipp, 2017; Reuter, Kaufhold, Spielhofer, & Hahne, 2017). In this regard, the promoted frameworks by the World Meteorological Organization for improving the response of communities during flood emergencies, such as local impact-based EWS, can benefit from these advantages hazard crisis apps can offer (Stanciugelu et al., 2017; WMO, 2021).

As part of the recently developed and implemented community-based "site-specific EWS" (SS-EWS) (Meléndez-Landaverde & Sempere-Torres, 2022), a flood crisis mobile application named "A4alerts" has been designed to disseminate the location-specific flood warnings and associated self-protection actions triggered by an operative impact-based EWS. In this context, the A4alerts was developed to support the rapid flood risk communication and help enable appropriate actions to decrease the negative impacts at vulnerable points or community areas. This work presents the A4alerts app, its functionalities and the first online user experience evaluation performed in Catalonia, Spain.

The paper is structured as follows: Section 2 provides background on the previous work regarding crisis mobile apps for emergency communication. Section 3 presents the A4alerts, its functionalities and the operational EWS behind the app. Section 4 describes the experimental online demo developed for evaluating the A4alerts from an end-user perspective within the context of flood emergencies. Section 5 deals with the results and their implications. Section 6 describes the encountered limitations and areas for future research. Finally, some concluding remarks are presented in Section 7.

2 | BACKGROUND

An essential emergency feature of any effective flood EWS is the rapid dissemination of relevant information and behaviour advice to a broad audience of exposed and vulnerable individuals. As part of the four key elements of end-to-end EWSs, the dissemination and communication of warnings in real time are essential to enable timely and appropriate mitigation and self-protection actions to help reduce the negative impacts of flood events (WMO, 2018). In light of this, various types of hazard crisis apps (e.g., specialised or multi-hazard) with specific functionalities (e.g., citizen-generated reports) have been used in the past decade to facilitate the interactions and information to both authorities, responders and potentially affected end-users (Colombelli et al., 2020; Grinko et al., 2019). In Germany, the KATWARN and NINA apps developed by the Fraunhofer Institute (on behalf of the German public insurance companies) and the Federal Office for Civil Protection and Disaster Relief are among the most popular crisis apps officially available (Reuter, Kaufhold, & Knipp, 2017; Reuter, Kaufhold, Spielhofer, & Hahne, 2017). While they share similar functionalities, the mechanism to trigger and disseminate different types of warnings (e.g., weather-induced or for public events) varies and, thus, are encouraged to be jointly

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used during emergencies. In the United States, the FEMA app provides flood warnings, behavioural recommendations and enables users to report incidents through GPS photos. Previous qualitative studies comparing these three apps indicate that although they are, in general, well-perceived by the group of participants, their individual functionalities and strengths could be merged to develop an enhanced crisis app version to satisfy the different needs and requirements of potential end-users (Reuter, Kaufhold, & Knipp, 2017; Reuter, Kaufhold, Spielhofer, & Hahne, 2017).

Until now, the general development of hazard crisis apps and flood-specific technological platforms has focused on the technical capabilities and their implications regarding real-time implementation (Colombelli et al., 2020; Dallo & Marti, 2021). However, recognising the importance of employing people-centred methodologies, the work on hazard crisis apps is shifting from the traditional technocratic perspective. In this context, the latest research has focused on exploring how their design and individual features can effectively support the decision-making process of end-users to increase the probability of continuous usage and social integration (Paul et al., 2021). The usefulness of hazard crisis apps during emergencies from the point of view of potential end-users; the functional appropriateness of their features to support their decision-making process; and the technical implications on learnability and social adoption are elements that have been explored in recent years (Dallo & Marti, 2021; Fischer-Preßler et al., 2021). As recognised by the participants in the study performed by Tan et al. (2020b), end-users will not install or keep using a hazard crisis app without any apparent added value towards usefulness and appropriate support during emergencies. In this regard, previous studies have proposed guidelines and recommendations for designing general hazard crisis apps and their functionalities based on the expectations and expressed needs of participants through workshops and guided discussions. However, despite these existing studies, there is limited research on how their results can be applied explicitly to the design of crisis app prototypes for hydrometeorological hazards. Moreover, there is a larger significant knowledge gap regarding the specific functionalities end-users require to effectively support their preparedness and mitigation actions during weather-induced emergencies. Additional studies are necessary to identify the requirements of communities, authorities, and responders regarding innovative communication technologies to justify the functionalities that should be incorporated in crisis apps in the context of flood events.

Finally, these apps have inherent obstacles to wider acceptance, adoption, and usage. The long-term effectiveness of hazard crisis apps as a communication channel during emergencies is strongly influenced by the number of active users, as seen in the recent COVID-19 pandemic (Trang et al., 2020). Although the popularity and public knowledge of hazard crisis apps have grown, improving the level of integration of these specialised apps in society and maintaining a stable user base are some of the main challenges ICT and authorities currently face (Fischer-Preßler et al., 2021). Moreover, vulnerable groups such as individuals with disabilities (e.g., visual or hearing impairments) are at risk of being excluded or neglected when considering the design and technical requirements of hazard crisis apps (Paul et al., 2021). Interestingly, though the elderly have been categorised as technological-averse, the recent study by Appleby-Arnold et al. (2019) suggests these groups can have a positive attitude towards using hazard crisis apps to support their actions during emergencies. There is a demand for more focused and context-specific research on the factors influencing the acceptance and intention to use crisis apps during weather-induced events.

3 | THE A4ALERTS MOBILE APPLICATION

From 2020 to mid-2021, the A4alerts app was designed, developed and internally tested alongside a group of stakeholders ($n = 15$) integrated by the responsible civil protection authorities, first responders and local experts in disaster risk management of Terrassa, Spain, through the iterative community-based SS-EWS framework (Meléndez-Landaverde & Sempere-Torres, 2022). Figure 1a presents the six sections of the current A4alerts app: (1) main or home view; (2) site-specific warnings (SSWs); (3) report an event; (4) feedback; (5) official warnings; and (6) information. The description of the warning system behind the app, the functionalities and how the app works during flood emergencies are presented below.

3.1 | Operational background: The community-based SS-EWS framework

Developed at the Center of Applied Research in Hydrometeorology of the Polytechnic University of Catalonia within the H2020 ANYWHERE project,[1] the SS-EWS framework guides communities to co-design, implement and co-evaluate impact-based flood warnings (i.e., SSWs) alongside self-protection actions for specific vulnerable points (e.g., sites in a flood zone area) or community

[1]Source: https://tinyurl.com/mr2b4kmf.

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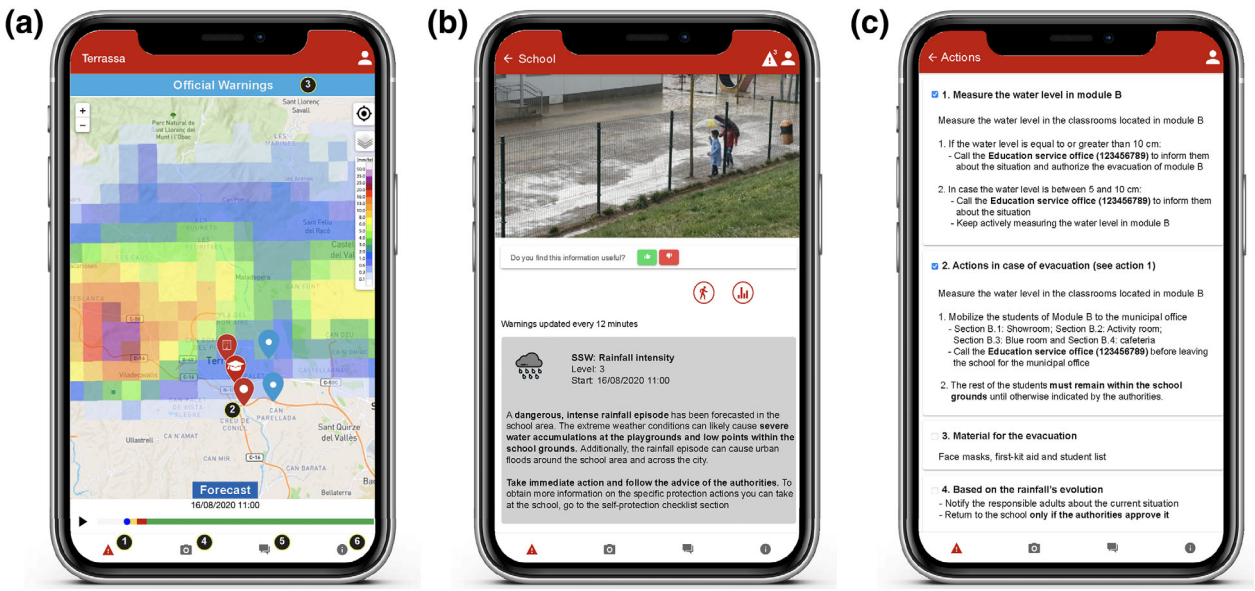


FIGURE 1 The A4alerts mobile app. (a) The home screen view (1) and its different sections: (2) active site-specific early warning system (SS-EWS); (3) official warnings; (4) report an event; (5) feedback; and (6) information. The sections are identified with a circle and the corresponding number. (b) The SS-EWS section. (c) The self-protection action checklist functionality.

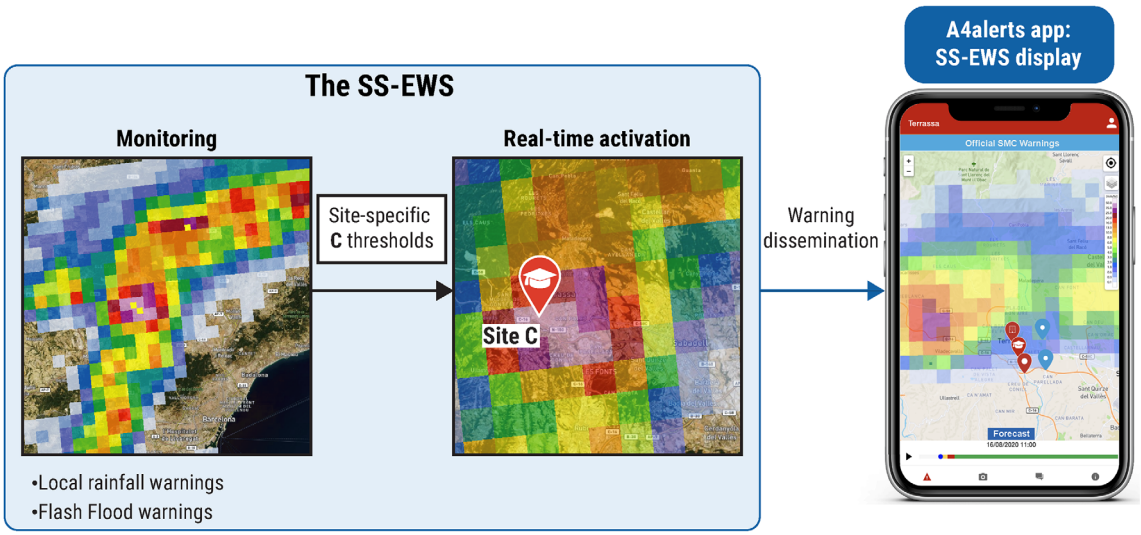


FIGURE 2 The A4alerts mobile app: The site-specific early warning system (SS-EWS) operational background.

sectors (e.g., transport). For this purpose, the method proposes steps to integrate hydrometeorological data (e.g., radar-based nowcasting and numerical weather prediction models) with local vulnerability, exposure and impact information to co-develop hazard-impact databases and establish site or community-specific impact-based rainfall thresholds. As seen in Figure 2, these are applied to trigger the two types of grid-based warnings within the EWS system, (i) local rainfall accumulations and (ii) flash flood warnings using the basin-aggregated rainfall concept (Corral et al., 2019). The first warning deals with the rainfall accumulated within a fixed duration at all the neighbouring pixels (1000 m) draining to a specific area. Similarly, the second one assumes the flash flood hazard in a pixel (200 m) can be characterised by the rainfall accumulated upstream (at a duration equal to the concentration time of its basin) and expressed in terms of the exceeded return period. Although these two types of warnings do not consider urban drainage, they have proven to be fast and suitable for alerting pluvial and flash floods driven by intense precipitation (Versini et al., 2014). The final warning level of a site or community is then determined by the maximum level between the two warnings or by applying a set of

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logical operators (e.g., AND). More detailed information regarding the impact-based and site-specific EWS, the overall co-design framework and its evaluation can be found in Meléndez-Landaverde and Sempere-Torres (2022) as part of the series on operational impact-based EWS.

In the context of the SS-EWS, the A4alerts app has been created to contribute to the third and fourth key pillars of end-to-end EWSs, (iii) dissemination and communication; and (iv) preparedness at all levels. Consequently, the A4alerts works exclusively as the mobile-based dissemination platform and display for the triggered impact-based warnings while the operational site-specific EWS performs the real-time monitoring and implementation of the logical rules.

3.2 | A4alerts functionalities

Table 1 shows the 10 functionalities implemented in the current A4alerts prototype version. These have been selected based on the recent studies on the needs and requirements of end-users for the development of hazard crisis apps (Dallo & Marti, 2021; Tan et al., 2020a). In this regard, the A4alerts functionalities have the objective of (i) informing about the active site-specific and official rainfall warnings; (ii) proposing actions (per warning level) from official self-protection plans to mitigate local impacts; (iii) helping enhance flood risk understanding; and (iv) enabling the participation of end-users to support authorities, emergency responders and citizens.

Due to the complexity of the required validation, quality control and privacy measures for their widespread public implementation (Imran et al., 2015; Reuter & Kaufhold, 2018), the "report and event" and "photo validation" functionalities have been included in the A4alerts app only as prototypes to exemplify their usage. This decision was made to concentrate the initial efforts on exploring and understanding the uptake of these functionalities by the testing group and the participants of the online demo within the context of a flood scenario. Thus, these sections can always be accessed from the main screen. In the report an event section, end-users can notify about an incident they have observed by uploading an image and a short description to improve the understanding of the situation or facilitate its location. After manually validating them through the back-end system, these images are shown on the background map of the main screen for other testing users to see and verify. Finally, since the A4alerts is currently in constant revision, a feedback section has been introduced to allow participants to send suggestions or report any problems they have experienced while using the app.

TABLE 1 The 10 functionalities implemented in the A4alerts app.

Functionalities of the A4alerts app	Description
Impact-based warnings for specific sites (i.e., SSWs)	Local and context-specific alerts for vulnerable locations or community sectors. Includes information about the hazard and the potential local impacts
Official warnings	Overview of the active official warnings in the region
Rainfall map visualisation	Radar-based (up to 2 h) and NWP (up to 24 h) rainfall field map
Self-protection action checklist	Local and context-specific actions coming from self-protection plans users can perform at the sites during the emergency
Background map visualisation	Interactive map with different layers, which allows end-users to select and display specific hazard or emergency information
Report an event	Allows users to report local damages or incidents through geo-located photos and text messages.
Photo validation	User can validate the reports submitted by others.
Feedback	Users can send comments, suggestions, or report errors about the app and its performance
Warning history timeline	A list of the previous active warnings at specific locations or community sectors.
Images of previous events	Historical analogies through photos of past events with equal warning levels

3.3 | Active emergency mode: Local impact-based flood warning dissemination

When a SSW is triggered, the A4alerts app starts working in active mode and notifies end-users with a push notification on their phones (i.e. pop-up message). In the "main section" of the app (see Figure 1a), all the active sites are displayed with a previously chosen icon and a colour representing the severity of the triggered warning levels (e.g., orange for SSW level 2). By tapping these icons, end-users can access the "SSW section" (Figure 1b), consisting of the co-designed impact-based flood warning text for the site, pictures of relevant historical flood events in the area, the self-protection actions checklist, and the warning

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history timeline. For the last two, separate sections (accessible through individual icons) are available to monitor the development of the emergency and review the actions for the current active warning level (Figure 1c). When no warnings are active, the action checklist section displays the complete flood self-protection plan per warning level to help increase the awareness and familiarisation of end-users with these actions. Furthermore, if a warning from the authorities is issued, the main section of the app will also display a banner on the top of the screen to access the "official warning" section. Within it, end-users can review the latest warnings by the authorities, their duration, and any official weather information.

Finally, the main screen will show the evolution of the event by displaying the rainfall map from the radar-based nowcasting (for the first 2 h) and the numerical weather prediction models (up to 24 h).

4 | A4ALERTS USER EXPERIENCE EVALUATION

This study aims to evaluate the current version of the A4alerts app from an end-user perspective within the context of flood emergencies. Similar to previous studies (Meléndez-Landaverde et al., 2020; Terti et al., 2019), an experimental role-playing demo in an online environment was prepared to showcase the functionalities of the A4alerts app during a previous real-case flood scenario. For this purpose, the online demo was pre-tested by six members of the stakeholder group from Terrassa (see Section 3) and finalised based on their feedback. The last version of the demo was disseminated online via email within the internal communication network of first responders in Catalonia. The participants recruited on a voluntary (anonymous) basis included members of civil protection, emergency responders and local stakeholder groups of Terrassa and Catalonia.

4.1 | Materials: Online experimental demo

The five sections of the online demo are presented in Table 2. The first section consisted of general instructions and an introductory video to help explain the A4alerts app and its functionalities to the participants. Section 2 contained the flood scenario alongside their triggered SSWs delivered through a web version of the A4alerts app. As in previous studies (Meléndez-Landaverde et al., 2020; Potter et al., 2018), the information presented in the flood scenario introduced participants to their assigned role, its responsibilities, and their decision-making obligations in case specific protection actions should be needed due to the active SSWs. In this regard, the warnings were provided within an interactive web version of the A4alerts app. This web version was separately prepared and embedded inside the experimental demo so participants could explore and understand how the app and its functionalities (see Table 1) would work during a flood scenario. Accordingly, the rainfall episode used in this study was based on the 16th of August high-impact event experienced in 2020 at Terrassa. Although short in duration (2 h), it caused surface water accumulations across Terrassa, flooding of underground parking, fallen trees and notable rescue activities by first responders (Amores et al., 2020; Consejo de Ministros, 2020). The SS-EWS was run with the corrected hydrometeorological data to produce the warnings presented in the web version of the A4alerts app.

Sections 3 and 4 were dedicated to evaluating the A4alerts app and its functionalities after being used by participants in Section 2. Previous studies and guidelines have proposed extensive lists of indicators and metrics for post-evaluating technological tools (Lewis, 1995; Tan et al., 2020b; Tullis & Albert, 2013). For the first section of the A4alerts evaluation, the criteria used in the ANYWHERE-H2020 project (Gebhardt et al., 2019) were applied to assess the value and benefits of EWS platforms from an end-user perspective. In this regard, the ANYWHERE co-evaluation approach is based on an adapted version of the "Socio-Economic Impact Assessment for Research Projects" methodology (Passani et al., 2014) and

TABLE 2 Online experimental demo sections.

Section	Description
1. Introduction and instructions	Demo guidelines and explanatory video of the A4alerts and its functionalities
2. Flood scenario and A4alerts app trial	Role-playing experimental flood scenario with a web version of the A4alerts app
3. A4alerts app and functionalities: Overall user evaluation	Assessment of the A4alerts app and its functionalities based on the ANYWHERE criteria through Likert-scale positive statements
4. A4alerts functionalities: Usefulness's ranking assessment	Ten A4alerts functionalities ranked from the most to the least useful in the context of flood emergencies by the participants
5. Participant's profile description	Demographic questions for profile characterisation

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the norms 9126-2/25010 for evaluating software quality from the International Organization for Standardization (ISO, 2003; ISO, 2011). Accordingly, Section 3 prompted participants to evaluate the A4alerts and its functionalities through positively framed statements linked to ANYWHERE sub-criteria on variables such as usefulness, functional appropriateness, and adoption (see Table 3). For this purpose, a 5-point Likert scale ranging from "I completely disagree" to "I completely agree" was implemented.

As mentioned in Section 2, past research has identified an array of general features and needs end-users (e.g., citizens and authorities) have for hazard crisis apps (Grinko et al., 2019; Kotthaus et al., 2016). However, it is essential to uncover the functionalities perceived as the most useful in the context of flood emergencies to (i) help redirect and justify the future work and resources for their development and real-time implementation and (ii) develop straightforward structures to facilitate the rapid usage of these apps during emergencies (Ahmad et al., 2017; Asikin-Garmager et al., 2022). Consequently, in Section 4, participants were asked to rank the functionalities from what they perceived to be the most (1st place) to the least useful (10th place) during a flood emergency. A video showcasing the previously explored functionalities in Section 1 was played on a loop to assist participants during this section. Finally, Section 5 consisted of questions related to the demographic characteristics of participants (e.g., age, gender).

4.2 | Procedure

To increase the probability of capturing a significant number of responses and due to the ongoing COVID-19 restrictions, the user experience evaluation was conceived in an online environment. In this context, the survey application Limesurvey was used to code and publish the experimental demo online and to collect the responses. After reading the initial instructions and viewing a video of the A4alerts app, participants were exposed to the flood scenario. In this section, participants had to use the web version of the A4alerts app to access the latest SSWs, local behavioural guidelines and the other available emergency functionalities to decide if self-protection actions were needed based on the provided information. At the end of the flood scenario, participants were directed to the post-evaluation to evaluate the A4alerts app and its functionalities based exclusively on their previous experience. Finally, general questions to identify the demographic characteristics of participants were asked as a last step of the online demo.

5 | RESULTS AND DISCUSSION

The results of Sections 3 and 4 of the experimental demo were analysed using the IBM SPSS Statistics software, version 28 (IBM, 2021). Descriptive statistics were calculated for the positively framed statements linked to the

TABLE 3 Statements linked to the ANYWHERE criteria for participants to evaluate the A4alerts app and its functionalities.

Criteria	Sub-criteria	Statement	Answer scale
Effects on emergency decisions, working routines, and daily decisions	Usefulness	The app can be useful during emergencies.	1 = strongly disagree 2 = disagree 3 = neither agree nor disagree 4 = agree 5 = strongly agree
		The app can be useful for first and second responders	
Effects on knowledge sharing and individual adoption	User's adoption and acceptance	If available, I would use this app in case of emergencies	
		If available, I would recommend this app to a friend or family member	
	Knowledge sharing	I would be willing to send photos of local impacts through the app	
Understandability	Functional appropriateness	The functionalities are well integrated to support my protection actions	
		The functionalities are easy to understand	
Learnability	Self-explanatory interface	The app could be used without previous training	
		The interface is intuitive	
Attractiveness	Interface aesthetics	The interface has an aesthetically pleasing appearance	

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ANYWHERE criteria of Section 3. For the usefulness ranking assessment in Section 4, the non-parametric Friedman's test was performed to establish whether there was a statistically significant mean difference in the whole set of functionalities ranked by participants. Moreover, the Bonferroni post hoc analysis with the Wilcoxon signed-rank tests was conducted to explore the existence of statistical mean differences in all the possible pairwise comparisons (45) of the ranked functionalities (Kim, 2015; Marino, 2018). In this evaluation, the critical significance values (ρ) were set at 0.05 and 0.001 (for the post hoc analysis with the Bonferroni correction). A total of 76 participants, the majority male (57%) in the range of 45–54 (53%), completed the A4alerts app online evaluation from January to March 2022. Of those that participated, 86% indicated they had been exposed to floods, and only 25% currently had a type of crisis or warning app installed on their phones.

5.1 | A4alerts: Overall user experience evaluation

Figure 3 shows the percentage of agreement with the statements regarding the A4alerts app and its functionalities presented in Table 1. In this article, answers equal or below the "disagree" level are considered negative evaluations; equal and above the "agree" level as positive evaluations; and "neutral" is a state of indecision or lack of convincing in participants and, thus, analysed by the authors in the context of the total percentage for each answer level. Most respondents positively evaluated the A4alerts app and its functionalities for most of the statements linked to the ANYWHERE co-evaluation criteria. In this context, many participants considered that the current version of the app is well integrated, useful, and has functionalities that are understandable enough to support the protection actions that might be needed during flood emergencies.

For the overall A4alerts app evaluation, a significant percentage of the participants agreed that it could be a useful tool during flood emergencies (97%) and that they would use or recommend the app to a friend or family member (93%). However, even though the interface of the app was considered intuitive (87%) and aesthetically pleasing (75%), a large percentage of respondents were not confident that someone could use the app without prior training or first external guidance (87% equal or below neutral level). As mentioned at the beginning of Section 4, the population exposed to the experimental demo included active members within the first and second responder groups and official authorities in Terrassa and Catalonia. Although the app was considered useful to assist these groups during flood emergencies (96%), the results suggest that to successfully use the app and its functionalities, resources (e.g., time, personnel, and material) should be allocated to provide training or demo sessions. Non-surprisingly, previous research has found

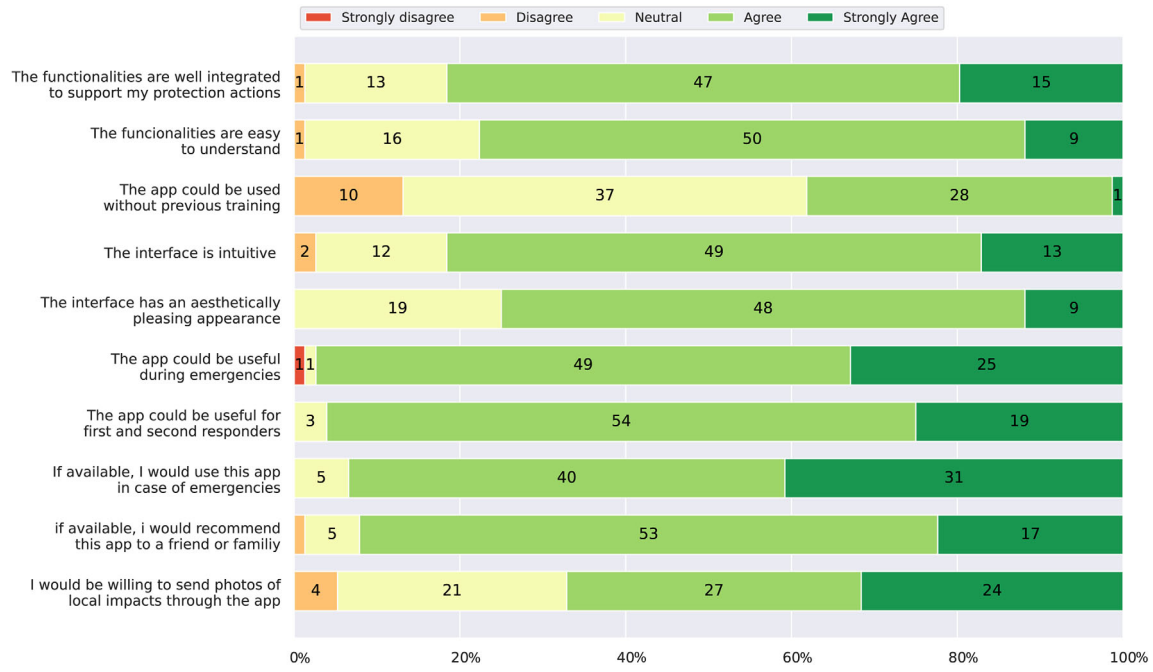


FIGURE 3 Percentage of participants agreeing with the statements linked to the ANYWHERE criteria about the A4alerts and its functionalities. Frequencies are given within the bars for each response type and statement.

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this factor as a barrier to broad adoption that new technology services commonly face (Dallo & Marti, 2021). Hazard and specialised crisis apps such as the A4alerts have specific design characteristics (e.g., warning colour codes and symbols); functionalities (e.g., radar maps, reporting and validation of impacts); and usability concepts that differ from the general domain of publicly available apps. (Fischer-Preßler et al., 2021). Moreover, irregular exposure to crisis apps emphasises the importance of providing resources and opportunities for straightforward handling and guidance (Reuter, Kaufhold, & Knipp, 2017; Reuter, Kaufhold, Spielhofer, & Hahne, 2017; Tan et al., 2020a). Thus, to facilitate the correct usage of the A4alerts app, it is necessary to consider learning methodologies (e.g., workshops, tutorials) to appropriately train end-users on the specialised flood emergency functionalities (WMO, 2021).

Regarding the functional appropriateness of the A4alerts features, many participants agreed that the individual functionalities were easy to understand (78%) and appropriately combined to support emergency actions during flood events (81%). As mentioned in Section 3, the functionalities incorporated in the A4alerts app follow the recommendations of recent qualitative research on the requirements and the preferences in features for disaster and hazard crisis apps by potential end-users (Dallo & Marti, 2021; Reuter, Kaufhold, & Knipp, 2017; Reuter, Kaufhold, Spielhofer, & Hahne, 2017). In this context, integrating functionalities and information desired by target end-users can help these technological emergency tools and services to be perceived as useful and, thus, increase the probability of adoption (i.e., installation and continuous use) (Fischer-Preßler et al., 2021; Grinko et al., 2019). The quantitative results of this study indicate that A4alerts app functionalities evaluated in the demo are positively perceived by participants to support their actions during flood emergencies.

Finally, due to growing research on innovative functionalities to support the contribution of end-users and enable their communication with emergency responders (Finazzi, 2020; Kaufhold et al., 2018), the willingness to send geo-located photos to report impacts was individually evaluated. In this study and as suggested by previous research (Appleby-Arnold et al., 2019), most participants agreed they would send geo-located impact photos through the app (67%). These features have been found to help enhance the situational awareness of authorities and citizens and enrich the local real-time information for emergency actions and future impact validations, a critical aspect of impact-based EWSs (Lara Tan et al., 2017; McCallum et al., 2016). However, their successful real-time implementation has some challenges. Data privacy and protection concerns can hinder the disposition of end-users to trust and share the personal and GPS information needed by these functionalities (Fischer et al., 2019; Yabe et al., 2022). Furthermore, awareness among users of the correct usage of these tools must be considered to avoid the spread of rumours and false information during flood emergencies (Alexander, 2013; Reuter & Kaufhold, 2018). In this regard, one possible step to overcome these shortcomings is recruiting local official volunteers. Limiting communication and sharing functionalities to these pre-identified groups can help improve the quality and reliability of collected data (Tavra et al., 2021). Additional concerns regarding their implementation include the need for advanced algorithms and skilled staff to correctly validate the information generated (Rossi et al., 2018).

Although the results suggest an overall positive evaluation, this should be seen as an initial exploratory assessment of the current A4alerts version and its functionalities within the context of flood events. Real-time evaluations must be performed to move from the perceived usefulness and intention of the participants to use the A4alerts app and its functionalities towards actual behaviour. Moreover, additional research is needed to determine the relationships and influence of variables such as individual functional appropriateness and usefulness of the A4alerts features with the actual adoption and acceptance rate of users. Nevertheless, the results for the A4alerts prototype and its functionalities are promising and serve as a base for validating future evaluations and development iterations to support the protective actions of end-users during flood emergencies.

5.2 | A4alerts functionalities: Mean ranks

Figure 4 shows the mean ranks of the 10 A4alerts functionalities evaluated in the fourth section of the online demo. The mean rank values are in descending order; hence, the highest-ranked functionality is closer to the value number one (i.e., first place). The results of the performed Friedman's test show an overall strong statistically significant difference (χ^2 (9) = 437.75; $\rho < 0.001$) in the mean ranks. However, the additional post hoc analysis reveals a non-significant difference between the four functionality combinations in Table 4. The implications of these results are discussed below.

The functionality ranked as the most useful out of the 10 implemented was the SSWs. This result agrees with recent studies where end-users have reiterated their need to receive more targeted and personalised emergency content (Kaltenberger et al., 2020; Potter et al., 2021). In this sense, these local warnings can represent a step forward in the impact-based flood warnings concept on the

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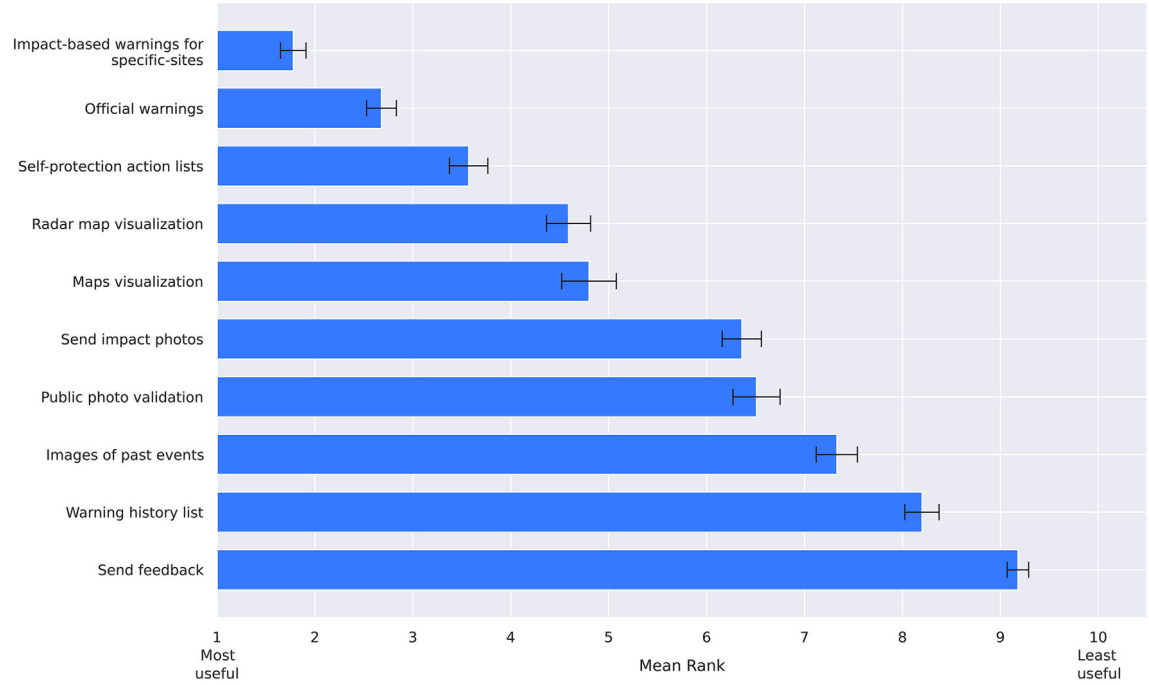


FIGURE 4 Mean ranks of the 10 evaluated A4lerts functionalities in descending order. The highest-ranked functionality is the one closer to the number one.

TABLE 4 Post hoc results for the non-significant pairwise functionality combinations of the usefulness's ranking assessment.

Functionality pair	Rank	*Z*-score	Sig (ρ)*
Self-protection action checklist Official warnings	Second	−2.788	0.005
Rainfall map visualisation Background map visualisation	Third	−0.868	0.385
Photo validation Report an event	Fourth	−0.419	0.675
Warning history timeline Images of previous events	Fifth	−2.871	0.004

Note: Sig (ρ)* < 0.001 with the Bonferroni adjustment.

assumption that they can help trigger specific protective responses and improve local flood warning comprehension (Taylor et al., 2019; WMO, 2021). The results of the work by Meléndez-Landaverde and Sempere-Torres (2022) suggest that end-users consider SSWs understandable and useful to empower vulnerable communities during emergencies. However, similar to impact-based warnings, the production and implementation of SSWs require significant resources and local information data that might not always be available (Harrison et al., 2021).

Due to their non-significant statistical mean difference, the following ranks (i.e., second to fifth place) are shared between the pairwise combinations in Table 4. For the second position, the official warnings and self-protection action checklist non-significant difference suggest that participants consider both equally useful and above the other seven functionalities implemented in the A4alerts app. Similarly, these two features have been identified in previous hazard crisis app studies as highly important for end-users in emergencies (Grinko et al., 2019; Karl et al., 2015). In the recent work by Dallo and Marti (2021), participants ($n = 25$) were questioned about their preference regarding information and useful functionalities within multi-hazard crisis apps. The need for real-time warnings alongside behavioural recommendations were the two most mentioned features perceived as relevant, congruent with the present evaluation results.

The non-statistical difference between the pairwise combinations sharing the third and fourth place can be explained by how they are frequently grouped based on their technical characteristics. The rainfall and information map visualisation have been categorised in previous studies as a unified feature of hazard crisis app requirements (e.g., "maps" or "interactive layers"). Similarly, sending photos of local flood impacts and validating them through the A4alerts app have been studied together within the concept of tools to support smartphone volunteering and citizen-to-authorities communication (Horstmann et al., 2018; Turpin & Holderness, 2015). In this study, these features are separated to explore and identify their individual level of usefulness inside these grouping categories and across the 10 implemented

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features for flood emergencies. However, the non-statistical difference indicates that these four functionalities were indeed perceived by participants as a group of app features and thus, resulting in equal levels of usefulness. In this regard, resources, design, and real-time implementation efforts should not consider them as separate functionalities but as part of larger thematic groups to effectively satisfy the requirements and needs of end-users for flood crisis apps.

The last two rank positions, fifth and sixth, are for the "images of past events and warning history timeline" pair and "send feedback" functionalities. Images and visual material have been suggested as an alternative to facilitate the digestion of flood warning information and to help increase the understanding of what the forecasted hydrometeorological parameters (e.g., intensities or return periods) mean based on previous experience in the area (Parker et al., 2009; Tan et al., 2020a). Furthermore, communicating the list of previous active warnings is a common practice in EWS to inform users of the evolution of weather-induced events (WMO, 2010). However, participants perceived both features as less useful than the other A4alerts features. Nevertheless, it is worth noting that this rank position is relative to the 10 implemented functionalities and does not imply that these are not useful during flood emergencies. Further studies are needed to identify the influence of flood impact images and any additional visual material on the risk perception and understanding of end-users in the context of ICT (Weyrich et al., 2019).

Finally, the sending feedback feature is the only non-exclusive functionality of hazard or flood crisis apps implemented and evaluated in the A4alerts. However, to constantly improve the behaviour of mobile apps, it is essential for developers to collect suggestions and comments from users (Martens & Maalej, 2019; Scherr et al., 2019). Moreover, prior research has identified that people tend to uninstall hazard crisis apps if they do not notice operational improvements in a long time (i.e., inactivity) or if too many handling errors are frequent (e.g., slow interface) (Tan et al., 2020b). Nonetheless, the relevance of this feature might only be apparent when errors are observed during real-time emergency usage and outside an online evaluation where users can directly communicate their feedback.

6 | LIMITATIONS AND FUTURE RESEARCH

As part of people-centred designs, user experience evaluations are essential to explore how individuals perceive and interact with technological tools (Asikin-Garmager et al., 2022). Nevertheless, due to the complexity of providing robust prototypes for these evaluations, previous research on crisis apps has used limited in-person experiences with low-fidelity prototypes or paper-based app demonstrations (Colombelli et al., 2020; Kaufhold et al., 2018). To further advance the field of hazard crisis apps, this study employed an operational and ready-to-be-used prototype of the A4alerts to evaluate the implemented functionalities from an end-user perspective.

In the context of the ongoing COVID-19 restrictions in early 2022, a strategic decision was made to conduct the evaluation within an online experimental environment, utilizing a web-based version of the app. This adjustment facilitated the broader dissemination of the demo within the different groups of stakeholders of Terrassa and Catalonia. However, the demo was not publicly accessible but exclusively distributed via email through the civil protection internal communication network of Catalonia. Consequently, the scope of responses was confined to these specific groups ($n = 76$). This limited reach, alongside inherent technological barriers (e.g., compatibility issues, stable internet access), could have resulted in fewer participants accessing and engaging with the demo. Moreover, the online experience limited the interaction with the participants to provide extra support or answer any questions related to the app or demo. Although a short video about the A4alerts was presented (in sections one and three of the demo), the restricted communication can also explain why most participants considered that additional guidance and training must be given to correctly use the app. Accordingly, additional workshops and in-person demonstrations should be conducted to build upon the existing findings. This approach will enable gathering first-hand experiences, interactions, and opinions from participants as they actively engage with the A4alerts application.

Finally, evaluating the A4alerts and its functionalities within the experimental demo is based on using behavioural intentions as a proxy for actual behaviours (Ajzen, 1985; Weyrich et al., 2020). Although a positive correlation between these two variables has been demonstrated in prior studies (Armitage & Conner, 2001), the intention to use a crisis app and its functionalities (e.g., sharing and validation) during emergencies can be influenced by several factors, such as trust in the system, risk perception and self-efficacy (Fischer et al., 2019). Real-time evaluation periods are needed to move from the initial exploratory user assessment of the A4alerts app and its functionalities to understand how it can be operationalised to support emergency responses and enable wider community acceptance. In this context, operational demonstrations of the A4alerts and its future iterations will be carried out in the next five years across municipalities in Catalonia (more than 20) and Europe

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(Portugal and Italy) within the framework of RESIST, a Horizon EU Mission Climate project.[2] These demonstrations will provide a notable platform for real-time evaluations of the A4alerts as an emergency communication tool. They will enable the assessment of the timeliness (i.e., lead-time) and accuracy of the warnings disseminated through the app and support its adaptation to regions characterized by diverse levels of vulnerability, social landscapes, and risk management governance structures. As mentioned previously, recognising that the level of social integration of technological tools influences their effectiveness, interactive e-learning activities and in-person workshops are being co-designed in an interdisciplinary and community-based environment (i.e., academics, social institutes, first responders) to appropriately engage, empower and train end-users while considering the requirements of the most vulnerable (e.g., elderly, hearing, and visually impair). It is foreseen that these long-term activities, alongside the operational demonstrations, will enable the A4alerts to evolve through iterative loops with the help of real-life experiences and user feedback. Furthermore, implementing the A4alerts in European regions with diverse risk profiles and climates offers the opportunity to explore the adaptation and transferability of the present findings towards developing crisis apps for other hydrometeorological hazards.

7 | CONCLUSIONS

Communities are demanding that risk information be quickly available and accessible, especially when the time window to perform local protection actions is limited. The widespread use of smartphones has opened the door for specialised hazard crisis apps to be a valuable complement to the rapid dissemination of flood warnings during emergencies. This research presents the A4alerts, an innovative flood-focused crisis mobile app developed to display the information from the community-based SS-EWS.

A first user experience evaluation of the A4alerts app and its functionalities within the context of flood scenarios was performed with Terrassa and Catalonia community representatives through an online experimental demo. The overall results suggest that the A4alerts was considered a potentially useful tool to effectively support users and their actions during flood emergencies. However, the results also indicate that resources should be allocated to provide supplementary training and guidance on how to successfully use the app. Furthermore, the functionalities implemented in the A4alerts were ranked by participants from the most to the least useful during emergencies. The SSWs, official warnings, and the self-protection action checklist were ranked as the most useful functionalities. Although innovative functionalities to enable the participation of end-users in providing and validating impact information were not among the first ranked, results suggest participants are willing to use them.

Understanding what and how to efficiently integrate technological applications and local needs for emergency communication is not a straightforward, one-way recipe. Nevertheless, the present evaluation of the A4alerts app can contribute to understanding the benefits, limitations, and areas of work within the field and as an example of good practice for other regions and communities. In this sense, the A4alerts app can have a supporting role in disseminating crucial local and impact-based warning information and breaching the communication gap between authorities, responders, and the most vulnerable communities during flood emergencies.

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DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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